



Effect of Forward Propeller Tilt and Inlet Shape in Propeller-in-Wing Thrust Vectoring

Grace G. Culpepper¹, Jielong Cai², and Sidaard Gunasekaran³

University of Dayton, Dayton, Ohio, 45469

We continue to investigate the efficacy of a propeller-in-wing thrust vectoring system utilizing a set of vanes placed in the wake of the propeller with two additional variables: the global forward tilt of the propeller with respect to the freestream and the inlet shape. All experiments were conducted at the University of Dayton Low-Speed Wind Tunnel (UD-LSWT) with off-the-shelf R/C electric propellers. Different configurations of propeller tilt angle and inlet shape geometry were tested considering a range of vane tilt angles. In comparison to the parallel orientation of the propeller disk, tested previously, the forward tilt of the propeller allowed for a wider operational range for vectoring thrust, itself achievable across a wider range of advance ratios. The circular duct-like wing opening also provided better performance than the square shaped wing opening in both forward thrust and pitching moment coefficient under the advance ratio range of vertical takeoff and for the transition to forward flight.

I. Nomenclature

C_D	=	Drag Coefficient	T	=	Thrust (N)
C_T	=	Propeller Thrust Coefficient; $C_T = \frac{F}{\rho n^2 D^4}$	V_∞	=	Freestream Velocity, (m/s)
C_Q	=	Propeller Torque Coefficient; $C_Q = \frac{P}{\rho n^3 D^5}$	θ	=	Propeller Tilt Angle (deg)
C_P	=	Propeller Power Coefficient; $C_P = 2\pi C_Q$	δ_v	=	Vane Tilt Angle (deg) δ_{ve} = Effective Wake Deflection (deg)
d	=	Propeller Diameter, (m)	γ	=	Propeller Pitch, (m)
J	=	Propeller Advance Ratio; $J = \frac{V_\infty}{nD}$	ρ	=	Air Density, (kg/m^3)
n	=	Rotational Speed per second	Subscripts:		
η	=	Propeller Efficiency/ Efficacy	H	=	Wing Horizontal Force
P	=	Power, (W)	V	=	Wing Vertical Force
Q	=	Torque (Nm)	X	=	Propeller Side Force
			Y	=	Propeller Normal Force
			Z	=	Propeller Axial Force

II. Introduction

Commercial interest in personal air vehicles (PAV) and unmanned air vehicles (UAV) with vertical takeoff and landing capability for air mobility transportation has grown significantly in recent years. Several propulsors have been considered as a primary source of thrust and lift in UAVs including fixed pitch multi-rotors, distributed electric propulsion, and wing-embedded propulsion systems. While the multi-rotor system offers superior hover performance, its efficiency in forward flight is relatively low when compared to a fixed-wing aircraft [1-2]. On the other hand, wing-embedded propulsion (WEP) systems have been widely considered as potential alternatives to multi-rotor systems

¹ Undergraduate Student, Department of Mechanical and Aerospace Engineering, AIAA Member. culpepperg1@udayton.edu

² Graduate Student, Department of Mechanical and Aerospace Engineering, AIAA Member. caij03@udayton.edu

³ Assistant Professor, Department of Mechanical and Aerospace Engineering, AIAA Member. gunasekarans1@udayton.edu

which provide higher cruise efficiency since their wings generate the majority of the lift as opposed to the rotors. It has also been shown that the local flow velocity on the wing upper surface induced by the embedded rotor augments the overall lift generated by the wing, thereby improving the aerodynamic efficiency [6].

The most common thrust vectoring mechanisms for WEP systems are rotor pitch control and collective blade pitch control. An alternative solution to the former set involves the use of a set of vanes positioned behind the propeller which redirects the high energy wake, thereby achieving thrust vectoring. The vanes may also reduce the swirl component in the propeller wake, increasing thrust efficiency [3-4]. Corgiat et al., Deckert, and Gerdes [5-7] investigated the performance of the Ryan XV-5, a jet-powered aircraft with WEP system technology, and concluded that the WEP system can reliably maintain gas-driven lift, while the performance in the transition to forward flight remains non-ideal due to a notably weak low-speed directional control response and complicated power management. This drawback could be improved upon using more modern flight control systems – which presents the opportunity to revisit and develop a thrust vectoring vane system for a modern PAV and UAV application.

Vaness have been used in the past to vector thrust from propulsors. Weir [8] used vanes to provide triaxial stability control of a free-flying ducted propeller. The results of the aforementioned study indicate a high effectiveness of the control vanes in influencing the system's pitching moment up to a 30-degree vane deflecting angle. Hickey and Ellis [9] investigated the performance of a semi-span wing featuring an embedded fan with both an inlet and exit. The duo concluded that exit vanes could be used to generate a net propulsive force for the attached system in a range of freestream velocities with only a small reduction in lift.

Our previous study [10] considered a propeller-vane in wing (PVW) system with the propeller disk placed parallel to the incoming freestream (0-degree tilt angle orientation). A set of vanes placed in the wake of the propeller was deflected from -30 to +30 degrees. Two configurations of propeller placement were investigated wherein the propeller disk was fully exposed to the freestream and then fully embedded in the wing. Experimental results for the normal/forward thrust coefficient and pitching moment coefficient for both these configurations are shown below in Figure 1. The results indicate that the system is capable of generating forward thrust for all advance ratios under $J = 0.28$ while maintaining the capability to generate both pitch-up and pitch-down moments under $J = 0.18$.

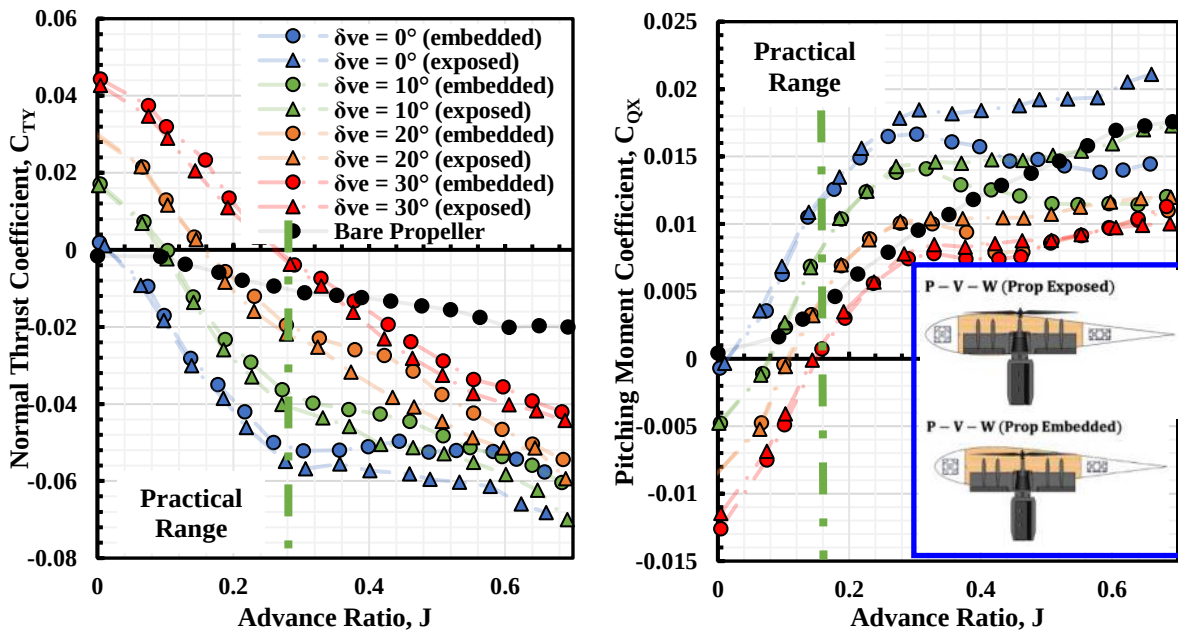


Figure 1. Propeller normal thrust coefficient (left) and pitching moment coefficient (right) across advance ratio conditions for different vane deflection angles in 0° tilt orientation.

Smoke flow visualization of the 0-degree tilt angle orientation for the PVW system reveals an unexpected streamline bending under propeller rotation. The freestream passes across the lower surface of the wing as expected for a standard airfoil geometry when the propeller is off. With the propeller on, however, the flow from the lower surface of the wing passes through the vane system and bends upwards, later pushed downward by the propeller once more, increasing the overall drag force experienced by the wing-rotor system. To avoid such flow bending and improve system efficacy, two possible solutions are proposed in this study: tilting of the propeller-vane system and modification of the square wing opening into a round, duct-like feature.

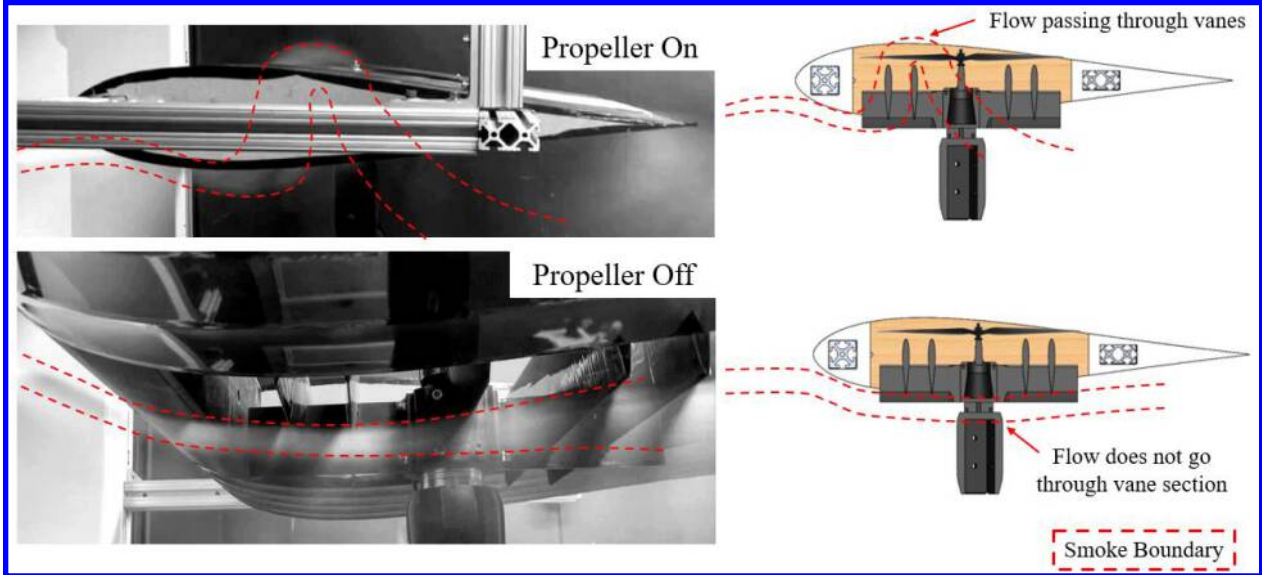


Figure 2. Side and underside views of flow visualization of PVW system (embedded propeller) at $\delta_v = 0^\circ$ and $J = 0.2$.

III. Experimental Setup

A. General Experimental Setup

Experiments were conducted on a thrust-stand at the University of Dayton Low-Speed Wind Tunnel Lab (UD-LSWT) in the open jet configuration. The UD-LSWT has a 16:1 contraction ratio, 6 anti-turbulence screens, and a 76.2 cm by 76.2 cm open jet test section. The effective length of the test section is 182.9 cm. A 111.8 cm by 111.8 cm collector accepts the expanded air on its return to the diffuser. The freestream turbulence intensity is 0.1% at 15 meters per second (as measured by hot-wire anemometry). A pitot tube connected to a TSI T600 Micromanometer was used to measure freestream velocity during the experiment.

The propeller-vane (PV) system was housed inside a wing with a span of 83.32 cm and a chord length of 60.96 cm. The wing section was constructed using an Eppler 479 airfoil, a design thick enough to house the propeller-vane system. A series of wooden ribs with the Eppler 479 profile were connected by both a leading edge and trailing edge spar made of 80-20 aluminum frame. The rib and spar structure of the wing was covered with a MonoKote [13] film, forming the skin of the wing. The fully integrated test setup is shown in Figure 3.

After construction, the wing was secured to the test section inlet by aluminum supports separate from those of the propeller-vane (PV) system (as shown in Figure 3b). In this way, the wing remained completely isolated from the PV system, itself mounted on a separate aluminum frame secured to the tunnel floor. Thus, force-based testing only measured forces on the propeller and vanes rather than forces acting on the PVW system as a whole. The initial design of the opening on the wing accommodates the PV system with the center of the propeller placed at the maximum thickness chord location (30% chord), thus providing better performance in lift generated by the wing [14]. To change the wing opening shape, the leading-edge piece and the trailing-edge piece of the wing can be replaced by different 3D-printed pieces while the remaining wing structure is unchanged.

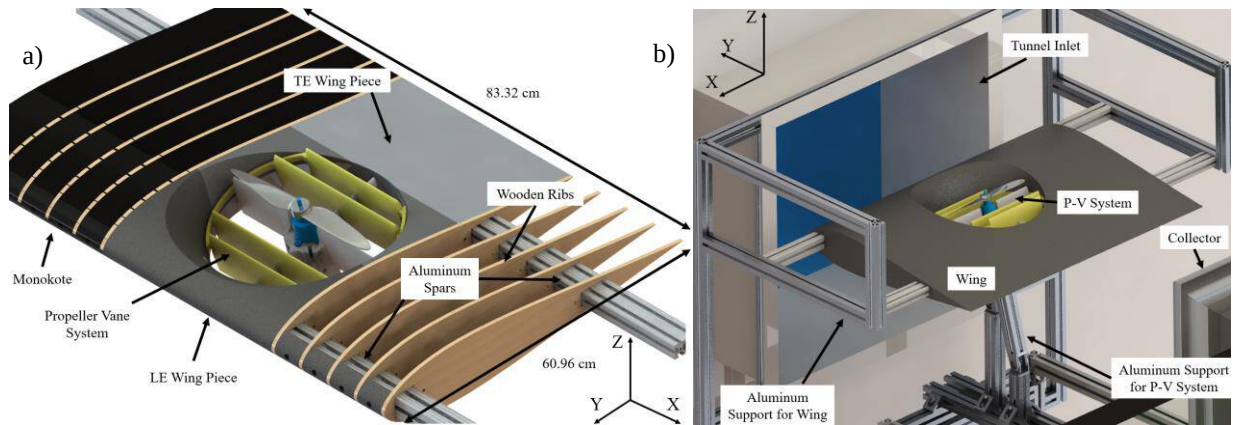


Figure 3. a) Internal assembly of propeller-in-wing test setup and b) isometric view of the propeller-in-wing test setup.

The APC 11-inch by 7-inch thin electric propeller was selected for testing because of the abundance of available results published with the same design selection from comparable literature data references [11]. Said propeller was driven by an OMA-3810-1050 brushless out-runner electric motor, itself powered by a PSW 30-108 constant-voltage supply. The peak Reynolds number of the propeller blades is 78,000 as calculated based on the blade chord at 70% blade semi-span and the peak propeller rotational speed achieved. The propeller rotational speed range, with its lower bound derived from signal-to-noise considerations and upper bound limited by the speed control capacity, is 60-120 RPS.

The propeller, motor and vanes were attached to an ATI Industrial Automation Mini-40 six-component force balance (www.ati-ia.com) [12]. Propeller thrust is defined to be the force acting along the Z-axis of the sensor while the propeller torque is the rotational moment about the Z-axis. These components are labeled in Figure 4 below. The sensor bolts to an aluminum-extrusion frame secured to the lab floor and wind tunnel contraction. The data sampling rate was 1,000 Hz for a 15-second duration at each testing condition, with two runs per each to improve data repeatability. In an effort to minimize sensor drift, tare values were taken before and after each test. The average of the two tares was then subtracted from the respective thrust and torque readings to correct for aforementioned drift. A band-pass filter was used to clean the data post-process, filtering out noisy frequencies related to propeller rotational speed and natural frequency of the thrust stand.

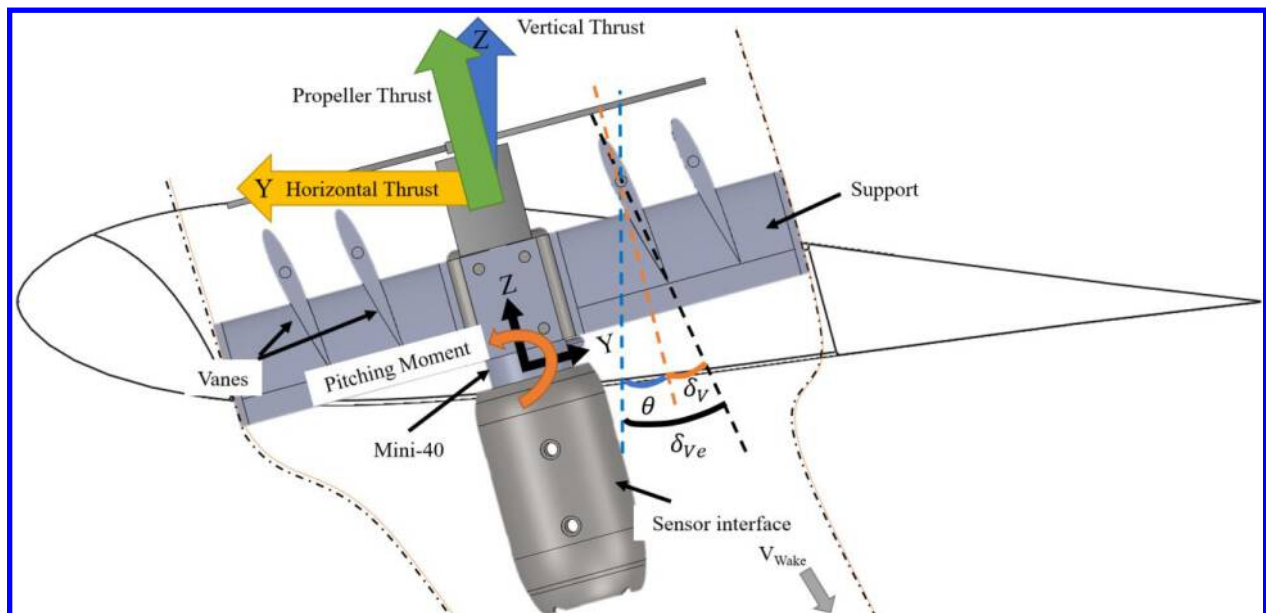


Figure 4. Sketch of test setup indicating the relationship between force components and propeller wake.

To examine the effects of forward propeller tilt angle (θ) on system performance, the entire propeller-vane (PV) system was oriented 15° forward into the freestream flow for a secondary test iteration. The location of the propeller hub, set at the upper surface of the wing, was maintained constant for both tilt conditions, matching that of the embedded propeller in 0° propeller tilt angle case. To compare the performance of the 0° tilt angle case to the forward tilt setup, the referenced coordinate system is set with respect to the wing, unpitched and constant in its placement. Thus, the resultant forces are shown as horizontal thrust T_H and vertical thrust (T_V) which can be written as follows:

$$T_H = T_Y \cos \theta + T_Z \sin \theta \quad (1.1)$$

$$T_V = T_Y \sin \theta + T_Z \cos \theta \quad (1.2)$$

Since the thrust-vectoring vanes are deflected from -20° to 20° (δ_V) for the 15° tilt angle (θ) case, to reliably compare experimental results from the 0° tilt case to that of the 15° , an effective wake deflection angle (δ_{Ve}) was introduced (shown in Figure 6), defined below:

$$\delta_{Ve} = \theta + \delta_V \quad (2)$$

Two different F/T sensor mounting configurations were developed. Both configurations are shown below in Figure 5. Configuration A (Figure 5a) measures the overall performance of the propeller-vane system, including both normal and axial thrust of the PV system as well as pitching moment. Since the vanes are acting as stators, however, the torque measurement from Configuration A does not represent the aerodynamic loading of the propeller. Configuration B (Figure 5b) directly measures the aerodynamic torque loading on the motor, which provides a direct measurement of the propeller power consumption.

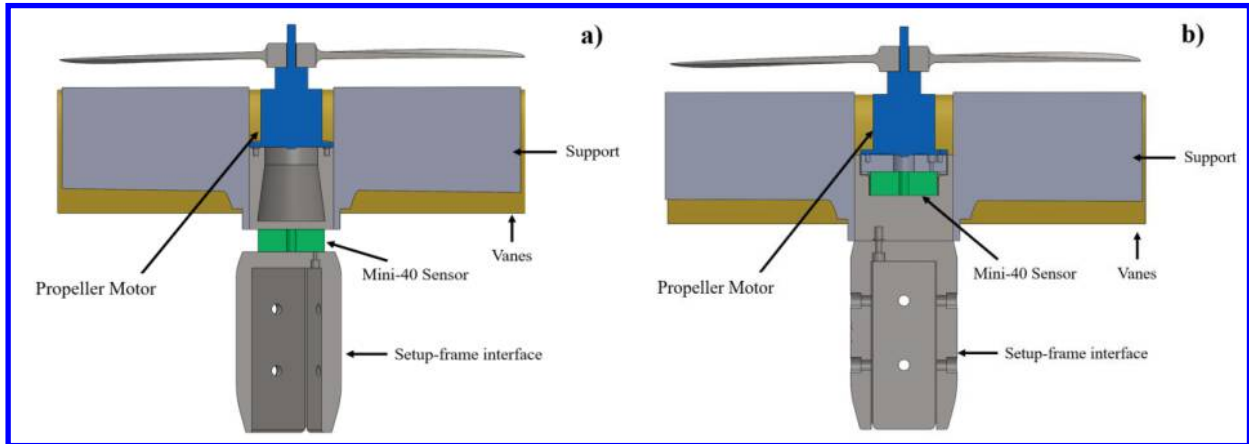


Figure 5. a) PV system direct measurement (Configuration A) and b) propeller direct measurement (Configuration B).

To examine the effect of the wing opening geometry on the propeller-vane system, a circular opening (Figure 6b) was designed with a lip length-to-width ratio of two, which has been found to provide better performance for a duct fan at high pitch/tilt angle ([8]-[9]). The size of the vanes employed in this new opening were reduced accordingly to compensate for the changes in the wing opening area.

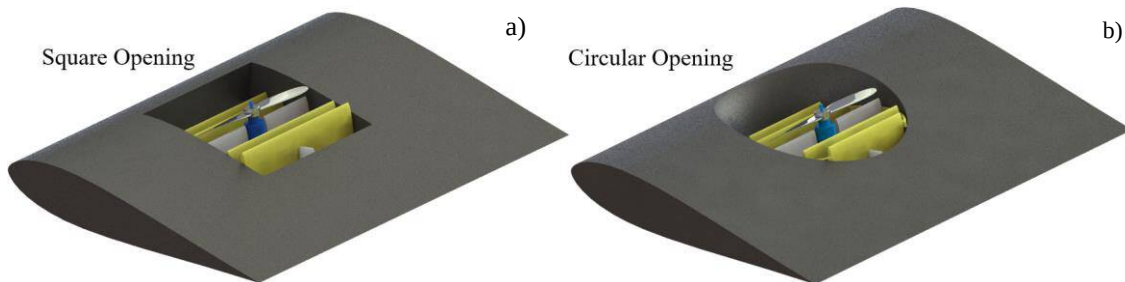


Figure 6. Isometric view for a) square wing opening b) circular wing opening.

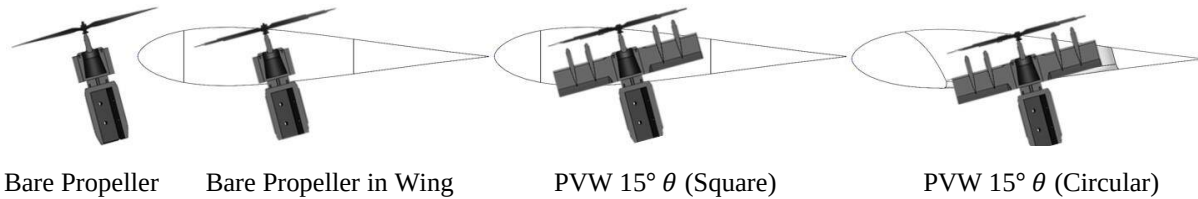
B. Test Matrix

The test matrix for this investigation is shown in Table 1. The performance of the bare propeller and bare propeller-in-wing are examined along with those of the PVW system cases. The propeller itself is tilted at a 15-degree tilt angle tested under both square and circular wing opening configurations. The propeller rotational speed is held at 100 RPS; thus, different advance ratios are obtained by changing the freestream velocity in the wind tunnel within the range of 0 – 19 m/s. It is worth noting that although the propeller disk is not placed perpendicular to the freestream, the calculations for the advance ratio follow that of a conventional puller-type configuration where the propeller disk is placed perpendicular to the freestream ($\theta = 90^\circ$).

As mentioned, the coordinate system for force measurement is described with respect to the wing orientation so that results remain comparable to the 0° tilt angle data from the previous study. For the bare propeller, since there is no wake deflection angle due to the lack of vanes, the effective wake deflection angle, δ_{ve} , will be the same as the propeller tilt angle, θ , according to Equation 2. For the PVW at a 15° tilt angle, the effective deflection will be the summation of θ and vane deflection angle, δ_v . For example, with a $20^\circ \delta_v$, the effective wake deflection angle δ_{ve} will be equal to 35° .

Table 1. Test Matrix.

Test Case	Inlet Shape	Propeller Rotational Speed (RPS)	Advance Ratio, J	Wind Tunnel Speed (m/s)	Effective Wake Deflection Angle δ_{ve} (degrees)
Bare Propeller	N/A	100	0 - 0.7	0 - 19	0, 5, 10, 15, 20, 25, 30, 35
Bare Propeller in Wing	Square				0, 5, 10, 15, 20, 25, 30, 35
PVW 15° Tilt (Configuration A & B)	Square				-5, 5, 15, 25, 35
	Circular				-5, 5, 15, 25, 35



IV. Results

Experimental results are discussed in three sections. The first section consists of results of the bare propeller and bare propeller-in-wing in order to investigate the effect of a wing on propeller performance without any special regard for vane placement. The second section consists of results from the 15-degree-tilt propeller-vane-in-wing (PVW) system orientation to study the effect of the propeller's forward pitch on the overall efficacy of the propeller-vane system. The third section consists of results of the duct-like wing opening tested in the 15-degree PVW system orientation to quantify the effect of duct opening geometry on the overall efficacy of the propeller-vane system.

A. Bare Propeller and Bare Propeller-in-Wing Performance

To understand the performance effects of wing integration on the propeller while establishing baseline cases for PVW system study, the performance of a bare propeller and bare propeller integrated into a wing (without the presence of vanes) are studied. The horizontal thrust coefficient C_{TH} for both cases is shown in Figure 7. A positive C_{TH} represents a forward thrust providing propulsive force for the system.

It is clear that C_{TH} for the bare propeller increases with the increment of advance ratio for the majority of the cases; when the propeller is integrated into the shrouding wing, this trend is reversed with a clear decrement occurring for all values at a higher advance ratio. Massive flow separation occurring at the wing opening as well as changes in the incoming freestream direction which the propeller encounters when integrated into the wing are likely the cause of such performance changes.

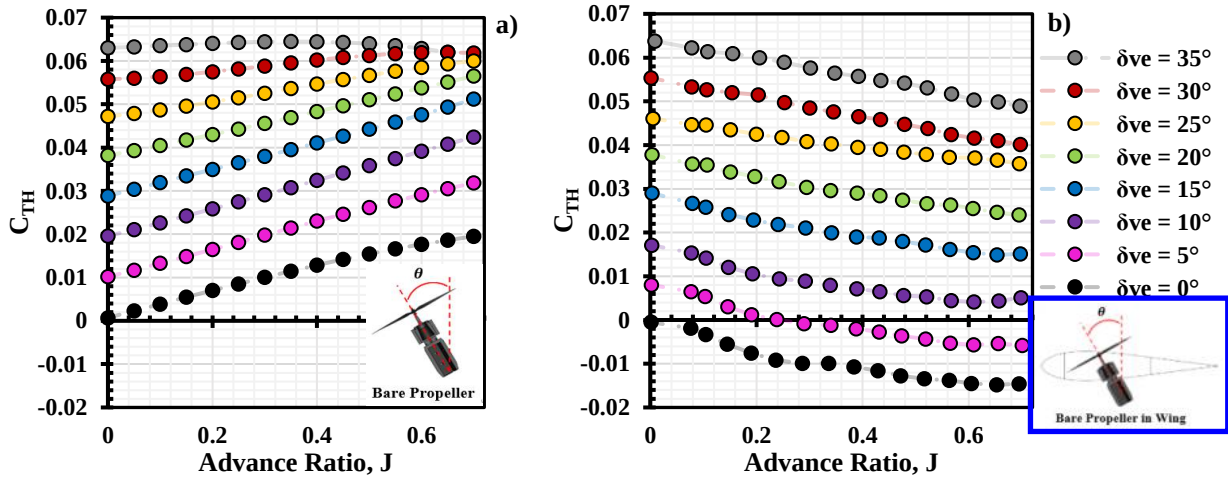


Figure 7. Horizontal thrust coefficient values across advance ratio conditions for a) a bare propeller and b) a bare propeller integrated in a wing with a straight square opening.

The vertical thrust coefficient C_{TV} is visualized in Figure 8: a positive C_{TV} represents vertical force that provides lift for the system in hovering and transition to forward flight. Both bare propeller cases show a very similar trend with a lower propeller wake deflection angle δ_{ve} resulting in a higher C_{TV} value at higher advance ratio. This trend reverses when increasing δ_{ve} . It is worth noting that the magnitude of the normal thrust, C_{TZ} , increases with the increment of δ_{ve} . When integrating the propeller into a wing, the changes in vertical thrust coefficient magnitude reduce, indicating a slightly positive effect of the wing on the propeller vertical thrust generation at higher δ_{ve} and a slightly negative effect at lower δ_{ve} . The overall magnitude of C_{TZ} remains similar when placing the propeller in the wing.

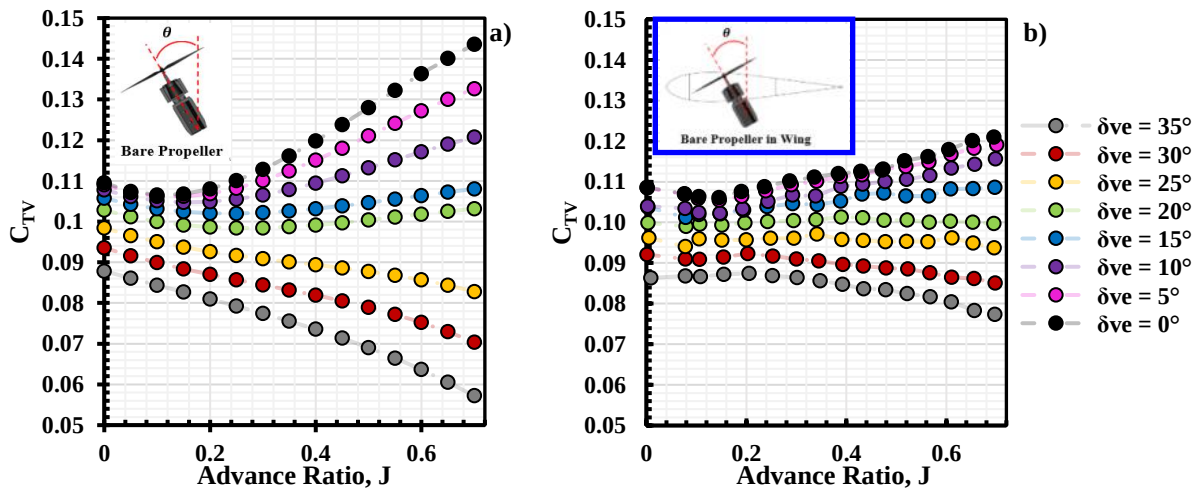


Figure 8. Vertical thrust coefficient values across advance ratio conditions for a) a bare propeller and b) a bare propeller integrated in wing with straight square opening.

Figure 9 shows the power coefficient of the tested propeller for both aforementioned configurations. The bare propeller has significant changes in C_p with the increment of J , and starts to deviate at $J > 0.4$ while the value of C_p for the bare propeller in wing almost remains constant throughout the J range tested, similar to the changes in C_{TV} seen in Figure 8. Again, a small deviation in C_p is observed at $J > 0.4$.

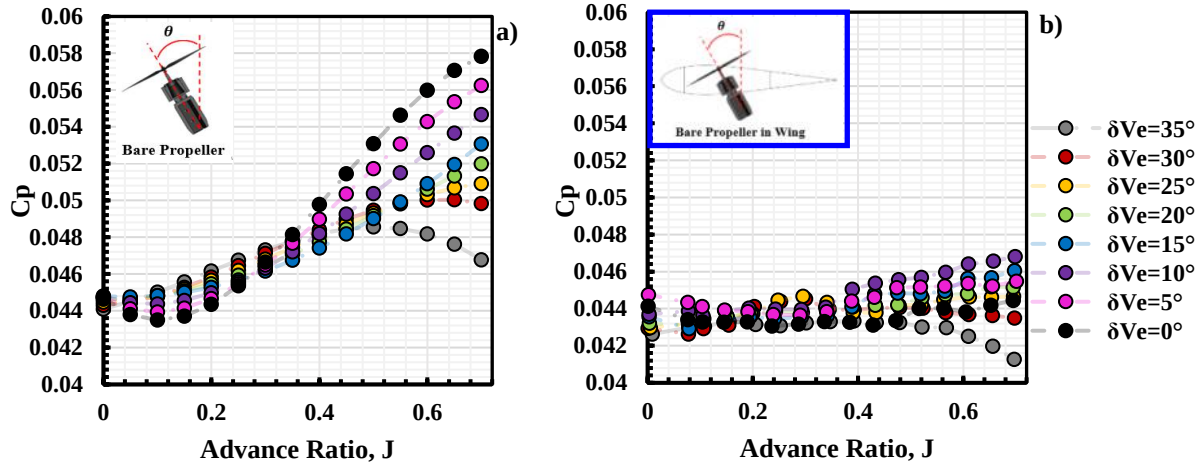


Figure 9. Power coefficient values across advance ratio conditions for a) a bare propeller and b) a bare propeller integrated in wing with straight square opening.

B. 15-Degree Tilt PVW System with Square Inlet

Figure 10a below shows the horizontal force coefficient C_{TH} for different effective vane deflection cases of the 15-degree tilt propeller-in-wing system at $0 < J < 0.7$, along with the data of the bare propeller at 15 degrees tilt angle. Again, a positive coefficient represents forward-vectored thrust generated by the deflection of the propeller wake, and a negative coefficient represents additional drag induced by the propeller-vane system. In general, with the increment of advance ratio, a clear decrement of C_{TH} is observed – a result which agrees with the overall trend seen in Figure 7b. A consistent difference in thrust vectoring performance with respect to vane angle occurs across advance ratios, with a higher deflection angle generating a more positive C_{TH} value. For $J < 0.7$, a positively directed normal force may be achieved with the maximum δ_{Ve} value. Moreover, the C_{TH} value for the $\delta_{Ve} = 35^\circ$ PVW case is higher than that of the bare propeller across all tested advance ratios. Recall from the C_{TH} plot for 0-degree tilt angle orientation in Figure 1, that C_{TH} will drop below 0 at $J = 0.3$. Deviation from this trend indicates an improvement in thrust-vectoring performance of the PVW system when tilting the propeller forward.

The behavior of the pitching moment coefficient C_{QX} for the PVW system is shown in Figure 10b. A negative C_{QX} indicates a nose down pitching moment and positive C_{QX} indicates a nose up pitching moment. When $J < 0.15$, both positive and negative values of C_{QX} may be achieved for the 15-degree tilt angle orientation of the PVW system by adjusting the vane deflection angle within the functional range of the design. When $J > 0.15$, however, additional control surfaces or small balancing rotors would be required to provide longitudinal stability for this system. Despite the increased range of successful thrust vectoring from this forward-pitched PVW setup as compared to the parallel-to-freestream propeller, the ability to provide longitudinal stability for the system remains limited to the same range as the 0-degree tilt angle case as seen in Figure 1b.

Figure 11 shows the vertical thrust coefficient C_{TV} and propeller power coefficient C_p for the 15-degree tilt angle orientation of the PVW system. Different from the trend seen in Figure 8, C_{TV} increases with the increment of advance ratio for all δ_{Ve} cases. Although greater axial thrust is produced by the propeller-in-wing system at a higher advance ratio, the design's ability to vector thrust diminishes significantly under these conditions while the power consumption of the propeller increases simultaneously. Higher vane deflection angle cases do consistently exhibit lower vertical thrust coefficients, attributed at least in part to subsequently increased vertical thrust values. The

performance of the $\delta_{Ve} = 15^\circ$ case has the highest C_{TV} value as the actual vane deflection angle δ_V is equal to zero. Following are the $\delta_{Ve} = 5^\circ$ and $\delta_{Ve} = 25^\circ$ cases which include a small loss in C_{TV} , and a more significant loss in C_{TZ} at $\delta_{Ve} = -5^\circ$ and $\delta_{Ve} = 35^\circ$. Again, an increment in C_{TH} corresponds to lower observed axial thrust. At $J = 0$, representing the case of vertical takeoff and hovering, a 10% loss in C_{TH} is observed for the tilted PVW system when compared to the bare propeller at the same tilt angle. This indicates that with a small loss in vertical takeoff performance, the 15-degree tilt angle orientation achieves better performance for thrust generation in forward flight.

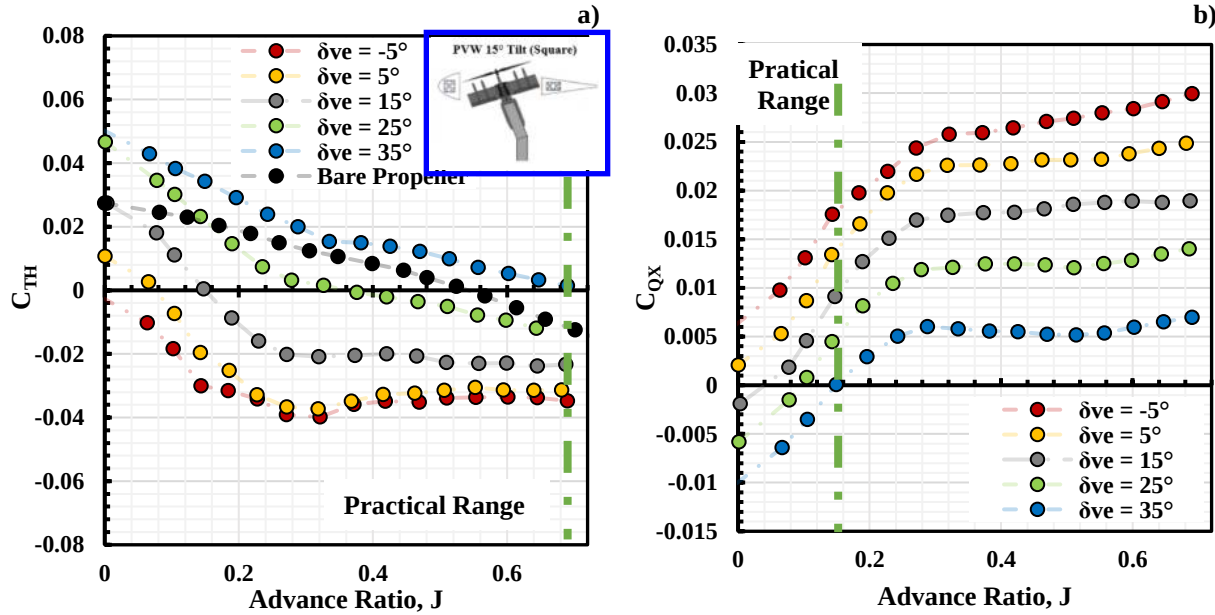


Figure 10. a) Horizontal thrust coefficient and b) pitching moment coefficient across advance ratio conditions for different vane deflection angles of the 15° tilt orientation (square opening).

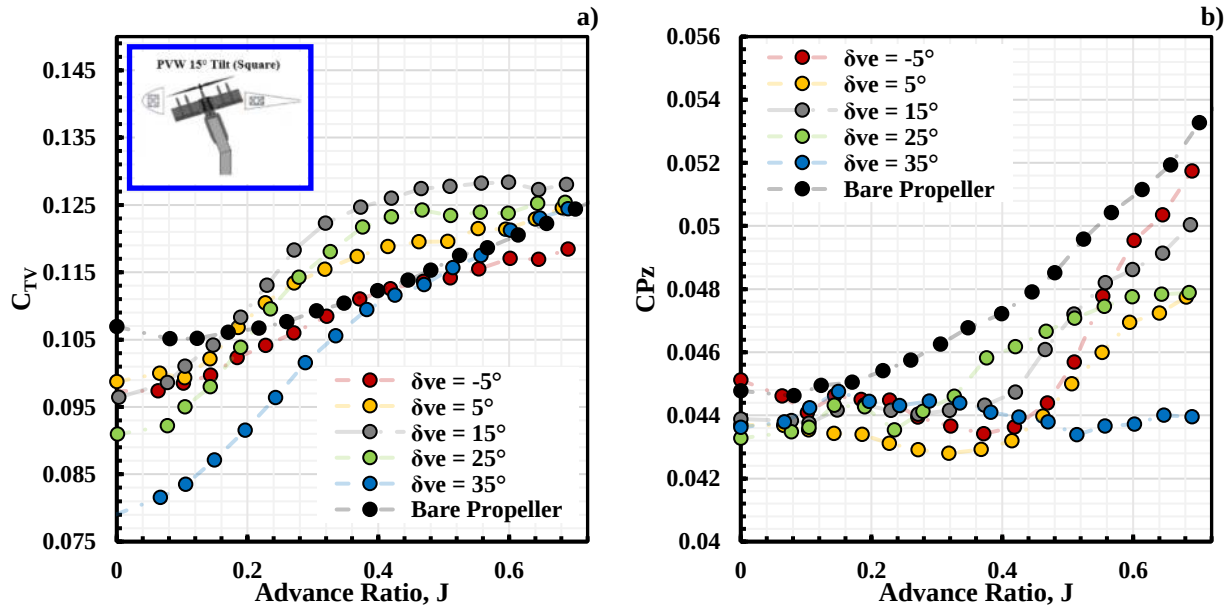


Figure 11. a) Vertical thrust coefficient and b) power coefficient across advance ratio conditions for different vane deflection angles of the 15° tilt orientation (square opening).

The difference in power coefficient, C_{Pz} , is small at low advance ratios, with the higher vane deflection cases, $\delta_{Ve} = -5^\circ$ and $\delta_{Ve} = 35^\circ$ experiencing a slightly higher C_{Pz} than other cases. The value of C_{Pz} remains relatively

constant for all cases until $J = 0.3$. At higher advance ratio conditions, an increment in C_{Pz} is observed with the increment of J , a trend also found in bare propeller performance. It is worth noticing that the C_{Pz} value for all PVW system cases is lower than that of the bare propeller at the same tilt angle, indicating a reduction in propeller blade drag when housing the rotor inside a wing, a result visible in Figure 9b above.

C. 15-Degree Tilt PVW System with Circular Opening

The following discusses results of the 15 degrees tilt angle orientation PV system in a wing featuring a circular, duct-like opening. Figure 12a shows the horizontal force coefficient, C_{TH} , of the design across advance ratio, J , at different effective vane deflection angles, δ_{ve} . When compared to the results from the square opening geometry shown in Figure 10, a small improvement in C_{TH} may be observed where $J < 0.2$, indicating improved performance for the PV system during the transition to forward flight. Here, it is worth notice that the total vane area for the circular wing opening is smaller than that of the square wing opening seen in Figure 6. This indicates that the extra vane area outside the projected propeller disk does not provide a performance benefit to the PV system. At $J > 0.3$, however, the square wing opening outperforms the duct-like opening, likely a byproduct of the extra vane area discussed previously. PVW performance for the $\delta_{ve} = 35^\circ$ case overlaps with that of the bare propeller at $J > 0.5$ which also represents the boundary for practical operation with respect to forward thrust vectoring. Again, the realistic advance ratio range for such application on a VTOL aircraft extends to $J \sim 0.4$. The performance at $J > 0.4$ is not the major focus for the study, we will only focus on the performance at $J < 0.4$ in the following discussion.

The pitching moment coefficient C_{Qx} (Figure 12a) follows the overall trend for the tilted PV system with a square wing opening as seen in Figure 10. It is worth noting that the PV system within the duct-like opening can generate upward and downward pitching moment up to $J = 0.2$, creating an operational range greater than that of the square opening design. This agrees with the improvement seen in normal thrust at low advance ratio conditions where propeller momentum dominates flow behavior for the same PVW geometry. Again, this indicates a significant impact from the shape of the wing opening on PV system performance at small J , precisely where the transition to forward flight occurs.

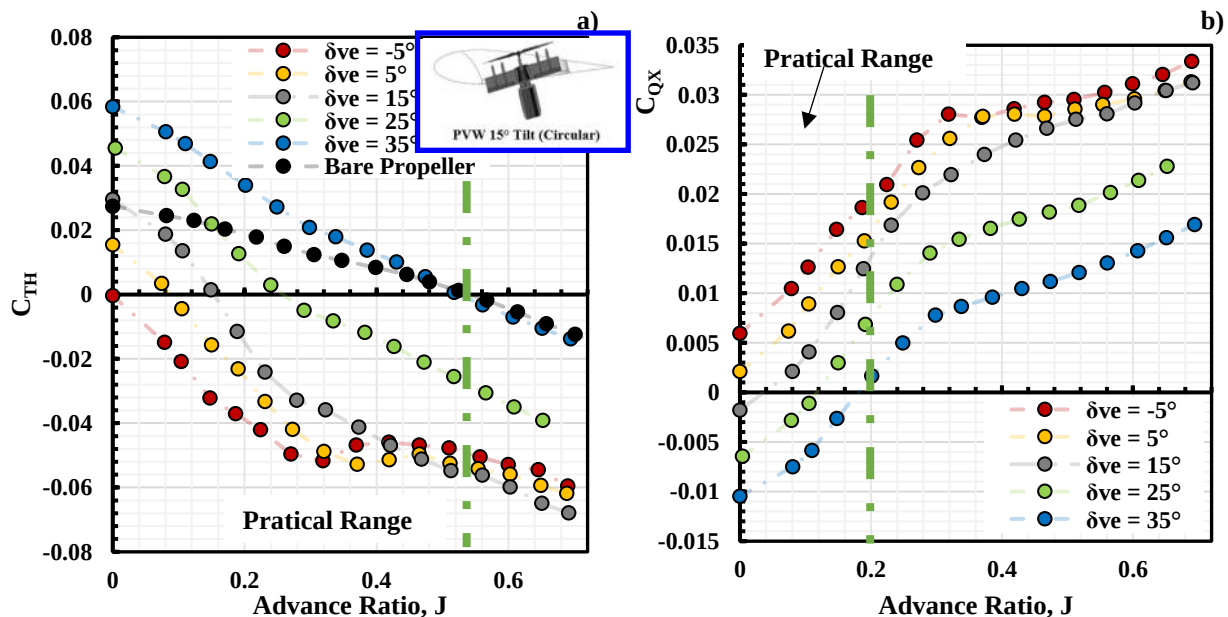


Figure 12. a) Normal thrust coefficient and b) pitching moment coefficient across advance ratio conditions for different vane deflection angles of the 15° tilt orientation (duct-like opening).

Results for vertical thrust coefficient, C_{TV} , which represents the vertical thrust generated for takeoff and landing, are shown in Figure 13a. The general trend observed in Figure 11 prevails, with a higher effective vane

deflection angle resulting a lower axial thrust for low advance ratios. At $J \sim 0$, the $\delta_{ve} = 5^\circ$ and $\delta_{ve} = 15^\circ$ cases of the circular wing design show a 4% increment in C_{TV} when compared to comparable square wing opening results. While for the power coefficient C_{Pz} (Figure 13b), the circular wing opening design outcomes resemble those of the square opening, as the changes are small until $J = 0.4$, which is outside the advance ratio range of interest. Overall, the circular wing opening provides a better performance for the PV system at $J \leq 0.2$ in both horizontal and vertical thrust coefficient as well as the pitching moment coefficient.

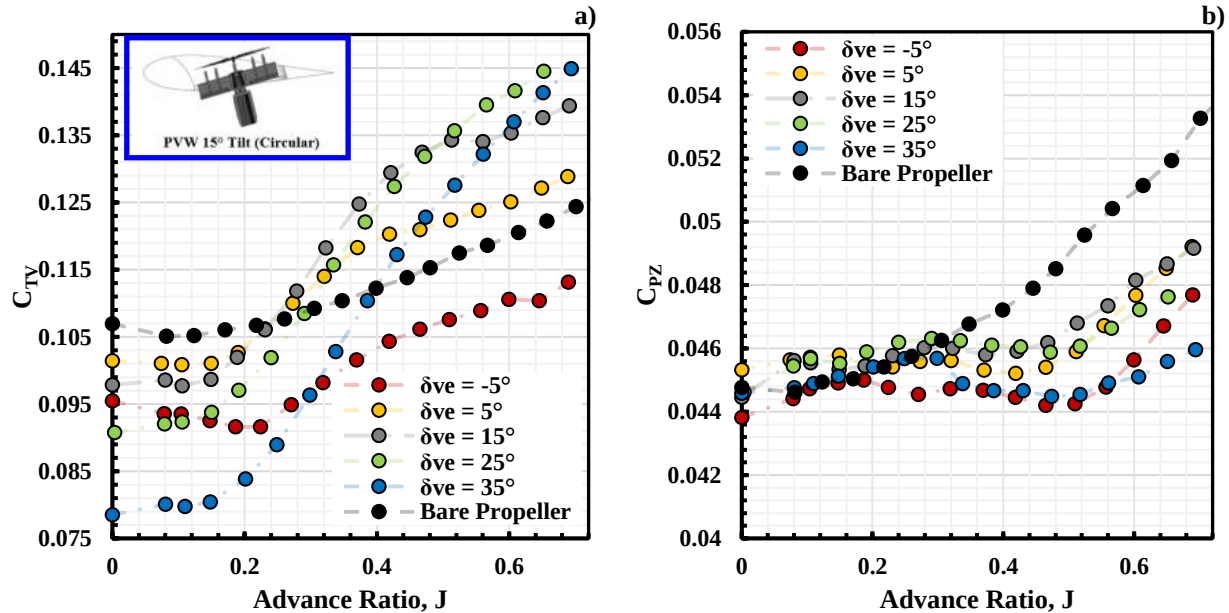


Figure 13. a) Axial thrust coefficient and b) power coefficient across advance ratio conditions for different vane deflection angles of the 15° tilt orientation (duct-like opening).

D. Smoke Flow Visualization

A smoke flow visualization is performed on the circular duct at propeller RPS of 100 and $J = 0.2$ to visualize the flow around the PVW system. From the flow visualization result on the square wing opening, shown in Figure 2, the rotation of the propeller induced an unexpected turning of the flow at the lower surface of the wing. This phenomenon also occurs for the circular wing opening, as shown in Figure 14. This bending of the stream tube in and out the circular duct will result in higher drag of the PVW system. Further investigation is required to fully understand the cause of such a flow turning phenomenon.

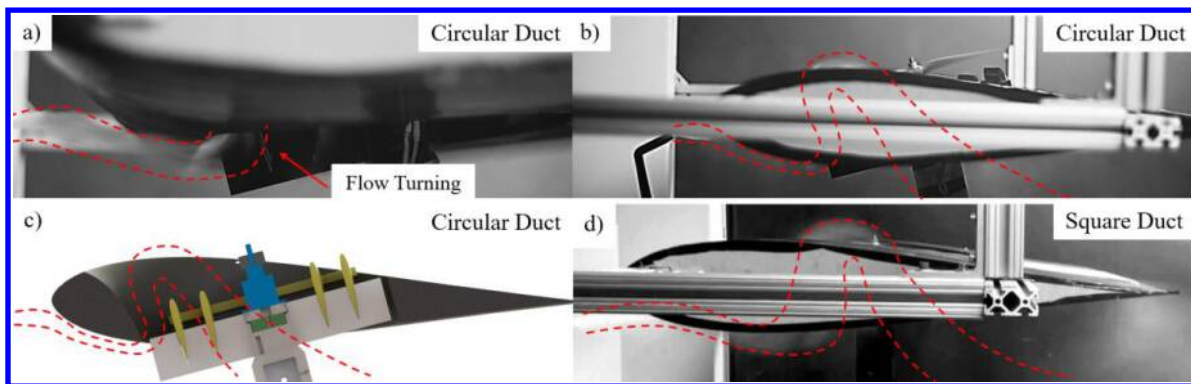


Figure 14. Smoke visualization on circular wing opening at 100 RPS and $J = 0.2$ with a) under-side view, b) side view, and c) cross-sectional side view schematic d) side view of square wing opening for comparison.

E. Overall Efficacy of the PVW System

The overall efficiency of the propeller-vane system in forward flight is defined by Equation 2. To reliably compare all combinations of propeller tilt and vane deflection, the system thrust efficacy, $\eta_{T\delta}$, is calculated using a bare propeller baseline case corresponding to the same effective vane deflection angle, δ_{Ve} .

$$\eta_{TV} = \eta_{T\delta} * \eta_p \quad (2)$$

where:

$$\eta_{T\delta} = \frac{C_{T\ PVW}}{C_{T\ Bare\ Prop}} \quad (3)$$

$$\eta_p = \frac{C_{P\ Bare\ Prop}}{C_{P\ PVW}} \quad (4)$$

Figure 15 shows the overall thrust vectoring efficiency for different vane deflection angles of the 0-degree and 15-degree tilt angle orientation cases of the PVW system with a square wing opening along with 15-degree tilt angle orientation cases employing the circular wing opening design. Experimental conditions leading to impractical system efficiencies ($\eta_{TV} < 0$) are excluded. The advance ratio is limited to $J < 0.4$ which represents the practical range of application for the PV system. In general, the system thrust vectoring efficiency decreases with increasing advance ratio in agreement with the trend established in Figure 1, Figure 10, and Figure 12. As the effective vane deflection angle increases, the practical advance ratio range of operation for the PVW system subsequently extends.

By tilting the propeller forward, the advance ratio range in which the PV system generates forward thrust extended. The 0-degree tilt angle orientation case only produces forward thrust where $J < 0.25$ while both 15-degree tilt angle orientation cases produce forward thrust up to $J = 0.4$ and higher advance ratio values. Regarding the effect of wing opening geometry, the PVW system using a circular opening produces an average of 10.5% higher overall efficacy as compared to the square opening for $J \leq 0.2$. This difference becomes smaller at higher advance ratios, though the duct-like wing opening still out-performs the square opening for $0.2 < J < 0.4$.

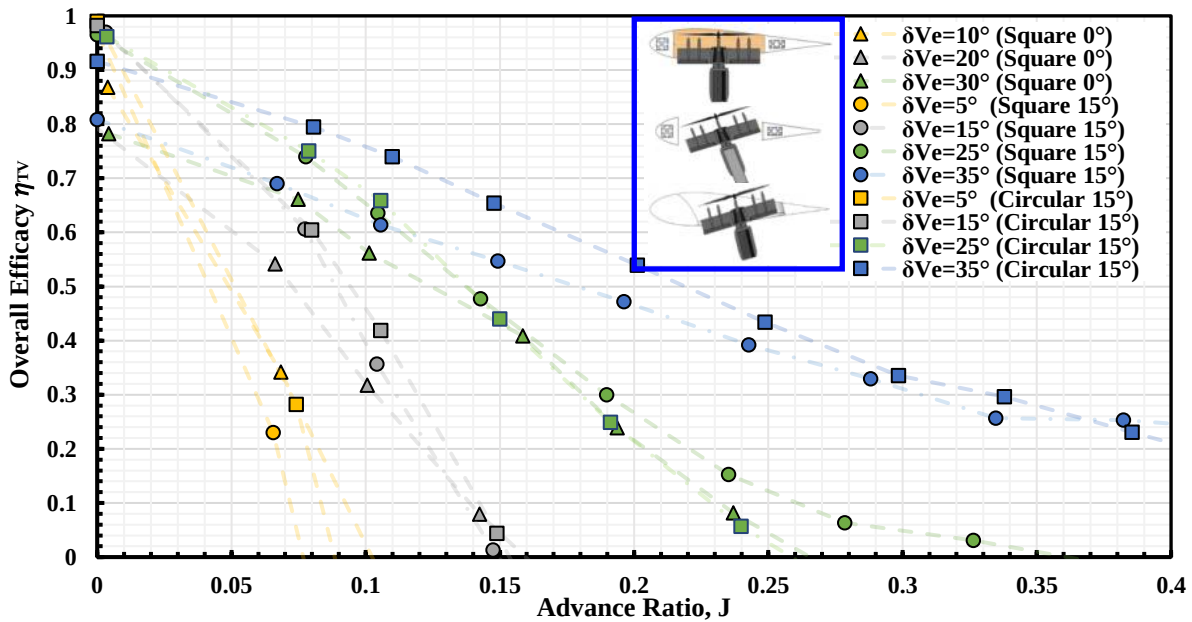


Figure 15. Overall thrust vectoring efficiency of all PVW iterations across advance ratio for different effective vane deflection cases.

V. Conclusions

In this study, the efficacy of thrust vectoring capabilities for the propeller-vane system embedded in a wing (PVW) was determined for different propeller tilt angles and wing opening configurations. The following are significant takeaways from testing results and analysis:

- For the 15° tilt PVW orientation, the system successfully produced forward thrust up to $J = 0.4$, showing significant performance improvement as compared to the 90° PVW system orientation.
- The circular duct-like opening provided improved performance in forward thrust and pitching moment generation for $J < 0.2$ as compared to the square opening geometry of the Eppler wing. This indicates that the wing opening shape has a significant impact on the PV system performance at small advance ratios where propeller momentum dominates flow behavior.
- The circular wing opening configuration achieved slightly better thrust vectoring performance with a lower vane surface area, also reducing the overall weight of the system.

VI. Acknowledgments

We would like to extend our thanks to the Clare Boothe Luce (CBL) program for supporting the work here completed, made possible by its continued support. The CBL program is committed to increasing the participation of women in STEM fields in higher education, and its commitment has been integral to the progression of this study.

VII. References

- [1] Stahl, P., Röbber, C., and Hornung, M., "Benefit Analysis and System Design Considerations for Drag Reduction of Inactive Hover Rotors on Electric Fixed-Wing VTOL Vehicles", AIAA Aviation, Atlanta, 2018, AIAA 2018-4150.
- [2] Stahl, P., Seren, T., Roessler, C., and Hornung, M., "Development and Performance Comparison of Optimized Electric Fixed-Wing VTOL UAV Configurations", ICAS 2018, Belo Horizonte, Brazil.
- [3] Ende, L.V.D., "Swirl Recovery Vanes for Propeller Propulsion Systems: An Aerodynamic and Aeroacoustics Investigation by Lattice Boltzmann Method", 2018, Delft University of Technology, Delft.
- [4] Stokkermans, T. C. A., "Design and Analysis of Swirl Recovery Vanes for an Isolated and a Wing Mounted Tractor Propeller", 2015, Delft University of Technology, Delft.
- [5] Corgiat, A., Lind, G., Hartsel, J., "Fan-in-Wing Technology, from the XV-5A to the Present". AIAA 1993.
- [6] Deckert, W., "The Lift-Fan Aircraft: Lessons Learned", 1995. NASA Ames Research Center, California.
- [7] Gerdes, R., "Lift-Fan Aircraft: Lessons Learned from XV-5 Flight Experience". AIAA 1993.
- [8] Weir, R., "Aerodynamic Design Considerations for a Free-flying Ducted Propeller", 1988. Sandia National Laboratories, New Mexico.
- [9] Hickey, D., and Ellis, D., "Wind-Tunnel Tests of a Semispan Wing with a Fan Rotating in the Plane of the Wing", 1959. NASA Ames Research Center, Washington.
- [10] Culpepper, G. G., Cai, J., and Gunasekaran, S., "Propeller and Propeller-in-Wing Thrust Vectoring" SciTech 2021, AIAA 2021-0615.
- [11] Brandt, J.B. and Selig, M.S., "Propeller Performance Data at Low Reynolds Numbers," 49th AIAA Aerospace Sciences Meeting, AIAA Paper 2011-1255, Orlando, FL, January 2011.
- [12] ATI, F/T Mini-40 Sensor, Retrieved from: <http://www.ati-ia.com/>
- [13] MonoKote film, retrieved from <https://www.hobbyzone.com/building-supplies/covering-related/covering/monokote/TOPQ0210.html>.

- [14] Norland, S., “An Investigation of the Lift Produced by a Fan in a Two-Dimensional Wing”. MIT, Massachusetts.
- [15] Pereira, J, L., “Hover and wind tunnel testing of shrouded rotors for improved micro air vehicle design”. University of Maryland, 2008.